

Project Planning

Project Name

Smart Lender – Loan Eligibility Prediction System

1. Introduction

Project planning is a critical phase that defines the roadmap for successful development and deployment of the Smart Lender system. This phase establishes project objectives, timelines, resource allocation, development methodology, and risk management strategies to ensure efficient execution.

The Smart Lender project focuses on developing a Machine Learning-based web application that predicts loan eligibility using applicant information and assists financial institutions in making faster and more accurate lending decisions.

2. Project Objectives

- Develop an intelligent loan eligibility prediction system.
 - Automate the loan approval process.
 - Improve prediction accuracy using Machine Learning.
 - Reduce manual verification efforts.
 - Build a responsive Flask-based web application.
 - Ensure maintainability and scalability.
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3. Project Scope

The project covers:

- Dataset collection and preprocessing
 - Exploratory Data Analysis (EDA)
 - Feature Engineering
 - Model Training
 - Model Evaluation
 - Flask Web Application Development
 - Integration of the trained model
 - Testing and Deployment
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4. Development Methodology

The Smart Lender project follows the Machine Learning Development Life Cycle (MLDLC).

Requirement Analysis



Project Design



Data Collection



Data Preprocessing



Model Training



Model Evaluation



Flask Application Development



Testing



Deployment

5. Project Timeline

Phase	Status
Requirement Analysis	Completed
Project Design	Completed
Dataset Preparation	Completed
Data Preprocessing	Completed
Model Training	Completed
Model Evaluation	Completed
Flask Development	Completed
Testing	Completed

Phase	Status
Documentation	In Progress

6. Resources Used

Software

- Python
- Flask
- Visual Studio Code
- Git
- GitHub

Libraries

- Pandas
- NumPy
- Scikit-learn
- Matplotlib
- Seaborn
- Imbalanced-learn (SMOTE)

7. Risk Management

Potential risks and mitigation strategies:

Risk	Mitigation
Poor data quality	Perform preprocessing and cleaning
Overfitting	Use cross-validation and model evaluation
Invalid user input	Implement input validation
Deployment issues	Thorough testing before deployment

8. Expected Deliverables

- Trained Machine Learning Model
- Flask Web Application
- Loan Eligibility Prediction System

- [Project Documentation](#)
 - [GitHub Repository](#)
 - [Demonstration Video](#)
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9. Conclusion

The project planning phase provides a structured roadmap for the successful development of the Smart Lender system. By following a well-defined plan, the project ensures timely completion, efficient resource utilization, and delivery of a reliable loan eligibility prediction application.